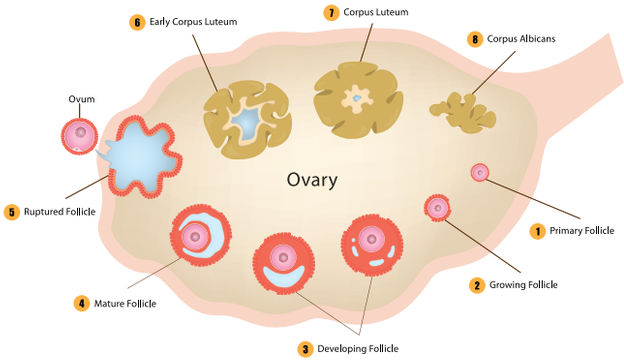


Figure 17.14 L.S of human ovary

17.8 Gametogenesis

The **formation of the sperm in male** and the **ovum in female** is called **gametogenesis**. It involves **spermatogenesis** (formation of spermatozoa) and **oogenesis** (the formation of ova). Gametes with **haploid cells** are produced through gametogenesis.

17.8.1 Structure of Human Sperm

The spermatozoan consists of head, a middle piece and tail. The **sperm head** is elongated and formed by the condensation of nucleus. The anterior portion has a cap like structure called **acrosome**. It contains **hyaluronidase** an enzyme that helps the sperm to enter the ovum during fertilization. A short neck connects the head and **middle piece** which comprises the **centrioles**.The middle piece contains the **mitochondria** which provides energy for the movement of tail. It brings about **sperm motility** which is essential for fertilization.

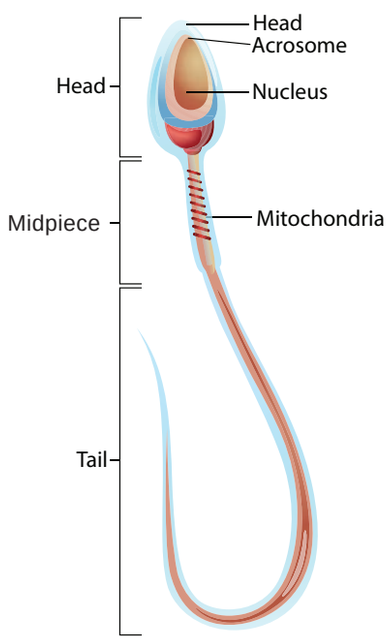


Figure 17.15 Structure of sperm

17.8.2 Structure of Human Ovum

The mature ovum or egg is spherical in shape.The ovum is almost free of yolk. It contains abundant cytoplasm and the nucleus. The ovum is surrounded by three membranes. The **plasma membrane** is surrounded by inner thin **zona pellucida** and an outer thick **corona radiata**. The corona radiata is formed of **follicle cells**. The membrane forming the surface layer of the ovum is called **vitelline membrane**. The fluid-filled space between zona pellucida and the surface of the egg is called **perivitelline space**.

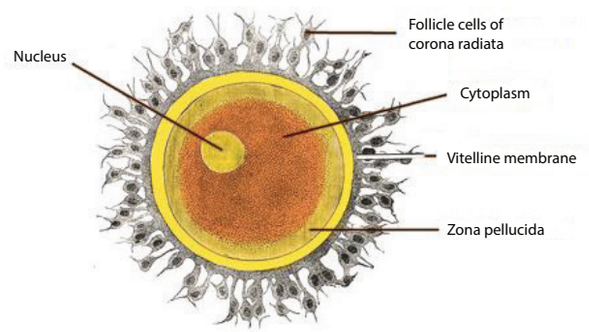


Figure 17.16 Structure of ovum

Puberty

The reproductive system in both males and females becomes functional and an increase in sex hormone production resulting in puberty. This phenomenon tends to start earlier in females than in males. Generally boys attain puberty between the age of 13 to 14 years, while girls reach puberty between 11 to 13 years. In male, the onset of puberty is triggered by the secretion of the hormone testosterone in the testes, in female the secretion of estrogens and progesterone from

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